DETECTING LITHIUM-6: A SPECTROSCOPIC REANALYSIS OF AN EXTREMELY RICH VERY METAL-POOR STAR

PROJECT COURSE - 1FA565

Master's Programme in Physics
May 2026

Cayetano Hernández Sánchez

Supervisors: Karin Lind and Ella Xi Wang



UPPSALA
UNIVERSITET

Contents

1 Summary **3**

2 Introduction **4**

3 Methodology **7**
3.1 Data reduction . . . 7
3.2 Computational environment . . . 7
3.2.1 Stagger . . . 7
3.2.2 Balder . . . 8
3.3 Optimization tests . . . 9
3.3.1 Resolution of the models . . . 10
3.3.2 Number of snapshots . . . 11
3.3.3 Initial abundance determination . . . 13
3.4 Determination of the abundance and the isotopic ratio . . . 14
3.4.1 Markov Chain Monte Carlo (MCMC) . . . 15
3.4.2 Non-extrapolated case . . . 17
3.4.3 Extrapolated case . . . 17

4 Results **19**
4.1 Subordinate line fitting . . . 19
4.2 Resonance line fitting: non-extrapolated case . . . 20
4.3 Resonance line fitting: extrapolated case . . . 22
4.4 Comparative Analysis . . . 24

5 Discussion **26**

6 Conclusions **30**

1. Summary

According to the standard big bang nucleosynthesis (BBN) prediction, ^{6}Li should be virtually nonexistent in old, very metal-poor (VMP) stars. However, a recent study by Sitnova et al. [2023] reported a detection of ^{6}Li in a VMP Li-rich giant star (Gaia DR3 ID = 1918529631627603072). This study employed 1D NLTE models with *a posteriori*, scalar 3D corrections and found an isotopic ratio of $\frac{^6\text{Li}}{^7\text{Li}} = 5.65^{+5.05}_{-2.51}\%$ and an abundance of $A(\text{Li}) = 3.42 \pm 0.07$. In this work, we reanalyze this star with self-consistent 3D NLTE models, which, unlike 1D NLTE models, capture the convective line asymmetry that may mimic the ^{6}Li signature. As a result, accounting for Bayesian statistics via a Markov Chain Monte Carlo (MCMC), we lower the isotopic ratio to $\frac{^6\text{Li}}{^7\text{Li}} = 2.519^{+0.619}_{-0.619}\%$ and also the abundance to $A(\text{Li}) = 3.186 \pm 0.026$. While this constitutes a statistical detection of ^{6}Li at a $\geq 2\sigma$ confidence level, a definitive confirmation still remains pending given current grid edge constraints and extrapolation limitations. This result highlights the importance of employing 3D NLTE models over alternative ones, such as the one from Sitnova et al. [2023], when tracing isotopic abundances in the context of Li-rich old, VMP stars.

Enligt den gängse förutsägelsen inom big bang-nukleosyntesen (BBN) borde ^{6}Li i praktiken vara obefintligt i gamla, mycket metallfattiga (VMP) stjärnor. En ny studie av Sitnova et al. [2023] rapporterade dock att ^{6}Li hade påvisats i en VMP-jätte av litiumrik typ (Gaia DR3 ID = 1918529631627603072). I denna studie användes 1D NLTE-modeller med *a posteriori* skalära 3D-korrigeringar, och man fann ett isotopförhållande på $\frac{^6\text{Li}}{^7\text{Li}} = 5.65^{+5.05}_{-2.51}\%$ samt en halt på $A(\text{Li}) = 3.42 \pm 0.07$. I detta arbete analyserar vi om denna stjärna med självkonsistenta 3D-NLTE-modeller, vilka, till skillnad från 1D-NLTE-modeller, fångar upp den konvektiva linjeasymmetrin som efterliknar ^{6}Li-signaturen. Genom att tillämpa bayesiansk statistik via en Markov Chain Monte Carlo (MCMC) sänker vi isotopförhållandet till $\frac{^6\text{Li}}{^7\text{Li}} = 2.519^{+0.619}_{-0.619}\%$ och även haltvärdet till $A(\text{Li}) = 3.186 \pm 0.026$. Även om detta utgör en statistisk detektion av ^{6}Li med en konfidensnivå på $\geq 2\sigma$, väntar en definitiv bekräftelse fortfarande på grund av nuvarande begränsningar i modellnätets kanter samt extrapolationsbegränsningar. Detta resultat belyser vikten av att använda 3D-NLTE-modeller framför alternativa modeller, såsom den från Sitnova et al. [2023], när man spårar isotopiska halter i sammanhanget av litiumrika gamla VMP-stjärnor.

2. Introduction

The nature of lithium is quite complex; it is one of the few elements produced during Big Bang Nucleosynthesis (BBN), and it can be produced by either cosmic ray spallation or stellar nucleosynthesis; both of which are, in principle, well-understood processes. However, the stellar production rate of lithium is still a matter of debate. Because of its fragility, lithium nuclei are depleted in stellar layers reaching $\sim 2.5 \times 10^6$K (Mucciarelli et al. [2019]), but that is only for the isotope ^{7}Li; for the more fragile ^{6}Li, this depletion occurs at an even lower temperature (Mott et al. [2017]). Additionally, the origins of both of them are known to be different; ^{7}Li is produced by the three already mentioned processes, whereas ^{6}Li is only produced by cosmic ray spallation, making it significantly less abundant (Prantzos [2012]). In this sense, historically, the value of $A(^7\text{Li}) = 2.0 - 2.2$ found in the old metal-poor dwarf stars in the Milky Way halo was thought to represent the primordial value of lithium. That conforms to what is known as the Spite plateau (Wang et al. [2022], Spite and Spite [1982]). However, precise measurements of the CMB by the WMAP and Planck missions determined a higher photon-baryon ratio of $\eta \approx 6.1 \times 10^{-10}$, and the primordial abundances measured with this ratio yielded a discrepancy of a factor of 3 - 4 between the primordial values and the observed ones in the Spite plateau. This is known as the first lithium problem. As proposed by Richard et al. [2005], Korn et al. [2006], and Fu et al. [2015], a possible solution for this problem is that depletion occurs via diffusion and turbulent mixing during their pre-main-sequence and main-sequence evolution. However, the exact origins of this turbulent mixing remain unknown, as well as the sub-Spite plateau lithium abundances of some very metal-poor stars (VMP) as discussed by Sbordone et al. [2010].

Furthermore, there is a second problem. In principle, in unevolved old VMP, ^{6}Li should be nearly non-existent according to predictions of cosmic ray spallation and standard BBN theory (Prantzos [2012]). The first process builds up ^{6}Li, but given the age of the stars in question, the contribution of spallation to their chemical composition should be negligible, and the second process fundamentally produces no ^{6}Li. However, there are several early claims of lithium detection in old, VMP main-sequence turnoff stars (MSTO) that are in direct contradiction with these predictions (Asplund et al. [2006], Smith et al. [1993]). Given the fragility of ^{6}Li, were these observations to be true, the probable depletion solution of the first lithium problem would imply an even greater depletion for ^{6}Li. This is known as the second lithium problem. In this sense, there are different known mechanisms for Li-enrichment, for example, there is the Cameron-Fowler mechanism (Cameron and Fowler [1971]), the only one known for ^{7}Li production, which can be triggered in multiple ways depending on the mass and the environment of the star. Among others, it can be activated by the Hot Bottom Burning

(HBB) occurring in intermediate-to-massive Asymptotic Giant Branch (AGB) stars, their thermal pulses, or Cool Bottom Processing (CBP), which occurs in lower-mass stars during the Red Giant Branch (RGB). Additionally, there is a possible production channel for ^{6}Li involving energetic phenomena on the surface of magnetically active metal-rich giants (Tatischeff and Thibaud [2007], Mott et al. [2017]). However, there is no known ^{6}Li enrichment process for evolved or unevolved VMP stars. What is more, in the case of a Li-rich giant VMP star, the first dredge-up and the RGB bump are known to deplete Li, so, in the case of a detection, it would be difficult to explain the presence of both isotopes.

Nonetheless, the mentioned early claims of ^{6}Li detection in very metal-poor MSTO stars employed classical 1D model-atmospheres which neglect convective processes, and which, with the exception of Asplund et al. [2006], employed radiative transfer computed by LTE. Regarding this, 1D models are static and cannot intrinsically account for convection, making use, instead, of mixing-length theory (MLT). On the other hand, 3D models solve the Navier-Stokes equations and naturally include convection, providing estimates for granulation and the exact velocity field per unit volume of the gas while also taking into account time variability. Furthermore, whereas LTE assumes that atomic populations are dictated exclusively by collisions, NLTE explicitly solves the statistical equilibrium equations and accounts for the radiation field present in the outer layers of the stellar atmosphere. As a result, NLTE models properly estimate the formation depth of the line in contrast to LTE ones, which overestimate it. Consequently, there are substantial differences when measuring the isotopic ratio with both 1D NLTE and 3D NLTE models.

The measurement of $\frac{^6\mathrm{Li}}{^7\mathrm{Li}}$ is a very complex process; it involves characterizing the small asymmetry that arises in the Li I 6707 Å resonance line due to the ^{6}Li presence. For that, apart from very high-quality spectra, models that account for 3D NLTE effects are needed. Velocity fields in 3D hydrodynamic stellar atmospheres (Cayrel et al. [2007], Asplund et al. [2006]) have been proven to generate an additional absorption in the red wing of the resonance line that mimics the ^{6}Li induced asymmetry. A feature that a 1D model cannot account for. Furthermore, the formation depth that LTE models overestimate directly affects the measurement of ^{6}Li. In this sense, Wang et al. [2022], Lind et al. [2013], and González Hernádez et al. [2019] already studied this kind of stars, accounting for 3D NLTE effects, and found no ^{6}Li in them. Still, recently, the study from Sitnova et al. [2023] reported a detection of ^{6}Li in a Li-rich VMP giant employing 1D NLTE models with scalar 3D corrections, being the only detection of this kind.

In this project, we aim to revisit this study, which, using data from Subaru, studied the star Gaia DR3 1918529631627603072 adopting stellar parameters $T_{\mathrm{eff}} = 4690 \pm 80$K,

$\log g = 1.34 \pm 0.13$ and [Fe/H] $= -2.43 \pm 0.07$ and found a ratio of $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = 5.65^{+5.05\%}_{-2.51\%}$ by fitting the Li I 6707 Å resonance line, and an abundance of $A(\mathrm{Li}) = 3.42 \pm 0.07$ by fitting the 6103 Å subordinate one. Our reanalysis accounts for state-of-the-art, 3D NLTE spectral synthesis models with self-consistent Li^7 modeling and aims to measure the isotopic ratio via a Markov Chain Monte Carlo (MCMC) fitting. The 3D model atmospheres employed here are from the Stagger grid (Magic et al. [2013]), whereas the NLTE radiative transfer code is Balder (Amarsi et al. [2016a], Amarsi et al. [2018]). In Section 2, the MCMC fitting of both lines and its constrains are detailed. In Section 3, the results for the best fit of both the isotopic ratio and abundance are shown; subsequently, we discuss what these best fits imply for the formation of Li-rich stars in Section 4. Lastly, in Section 5, some conclusions are provided.

3. Methodology

3.1. Data reduction

Following Sitnova et al. [2023], the observational data employed comes from the Subaru telescope (`https://smoka.nao.ac.jp/index.jsp`), specifically from the HDS (High Dispersion Spectrograph), and it was reduced with IRAF. HDS has a spectral resolution of $R = 45000$ and employs two different CCDs that have 2048 pixels (in the direction of the slit) by 4100 pixels (in the direction of light dispersion). The data was manually Theil-sen normalized and radial velocity corrected employing a Python script and was found to have an empirical S/N per pixel of 43.1 for the subordinate line and 52.4 for the resonance one, which is relatively low.

3.2. Computational environment

The computational calculations themselves were run on the Dardel cluster, of which NAISS (`https://www.naiss.se/`) is the responsible operator. As mentioned in the introduction, the 3D NLTE models employed are composed of Stagger for the 3D part and Balder for the NLTE one. This section summarizes in a succinct way the whole computational environment employed to run the simulations.

3.2.1. Stagger

Stagger (Magic et al. [2013]) is a grid of 3D of time-dependent, hydrodynamic model atmospheres of cool stars (`https://staggergrid.wordpress.com/`) which employs 'box in a star' geometry and solves the radiative hydrodynamics (RHD) compressible equations system taking into account time variability in the form of snapshots. It covers various evolutionary stages from the main sequence to turn-off stars to red-giant ones with models discretized according to the pertinent stellar parameters, that is:

- Effective temperature (T_{eff}): From 4000 to 7000 K.
- Surface gravity ($\log g$): From 1.5 to 5.0.
- Metallicity ([Fe/H]): From -4.0 to $+0.5$.

Interpolation between the synthetic spectra from these grid points is therefore necessary to generate models at more specific stellar parameters. In that sense, as mentioned in the introduction, the star of interest in this project has stellar parameters of $T_{\text{eff}} = 4690 \pm 80\ K$, $\log g = 1.34 \pm 0.13$ and $[\text{Fe/H}] = -2.43 \pm 0.07$ (Sitnova et al. [2023]). Subsequently, the closest points in the Stagger grid to our star are the ones corresponding to $T_{\text{eff}} = 4500$ K, $\log g = 1.5$ and $[\text{Fe/H}] = -2.0$ or $[\text{Fe/H}] = -3.0$. From now on, these would be referred to as `t45g15m2005` and `t45g15m3005`. However, it is important to note that, in Stagger, there are no models at $T_{\text{eff}} = 4500K$ with $\log g = 1.0$, nor models at $T_{\text{eff}} = 5000\ K$ with either $\log g = 1.0$ or $\log g = 1.5$. Because of that, the only models that we are able to use and perform interpolation with are, then, these two. Additionally, as already mentioned, Stagger models account for time evolution in the form of snapshots, of which both of our models have 10 different ones. Stagger models also account for two different resolutions, that is, 48x48x240 and 80x80x240, with the latter being, by definition, more precise but also more computationally expensive than the former one.

3.2.2. Balder

Balder is the NLTE radiative transfer code employed. Balder consists of two parts: Balder (Amarsi et al. [2018]) itself and Blue (Amarsi et al. [2016a], Amarsi et al. [2016b]). Balder is the radiative transfer code itself, whereas Blue is the accompanying opacity package, which is necessary to precompute the background line opacities. The execution of the code is primarily governed by four fundamental configuration files:

- **atmos.input:** This file links the radiative transfer code to the specific 3D hydrodynamic model atmospheres. It directs the code to the pertinent Stagger snapshots that provide the physical structure and velocity fields of the star.
- **multi3d.input:** This file controls the overarching execution parameters and convergence criteria. For the 3D synthesis, the input microturbulence (`vturb`) is strictly set to zero, as the macroscopic velocity fields are already natively modeled. The internal radiation field calculations are managed by `nmu`, which defines the number of polar angles that are used and `nphi`, which defines the azimuthal ones. Similarly, the emergent flux calculations for the spectrum are managed by `nmuout` and `nphiincout`.
- **atom.input:** This file designates the target element (Li) and specifies the paths to the required atomic models. Two different atomic models are used, one with an isotopic ratio of 0% (`atom.Li21_newcol_correct_0`) and another one with 6% (`atom.Li21_newcol_correct_6`). The specific atomic abundance is also explicitly set here (e.g., 3.42...), calculated following the relation $A(X) = \log_{10}\left(\frac{X}{\text{H}}\right) + 12$.

- **blue.input:** This file manages the background line opacities interpolated from the Blue outputs. The background abundances are appropriately scaled to match the stellar metallicity of the employed Stagger model (`abund.m200` or `abund.m300`). Notably, a `skip='LI';` parameter is implemented to omit the background line transitions of lithium. Furthermore, to prevent double-counting, the background opacities were strictly precalculated without the lithium contribution (`m200_noli.blue` or `m300_noli.blue`).

The appropriate configuration of these files allows us to run the `t45g15m2005` and `t45g15m3005` models for different abundances and isotopic ratios. Generating, as a result, among others, an angle-dependent specific intensities for both the total radiation field (`iet`) and the continuum (`iec`), a spectrum (`sp`), and a file that contains all the geometrical configurations of the calculus (`detailed_stagger`).

3.3. Optimization tests

Since 3D NLTE models are substantially expensive in computational terms. Before attempting any of the measurements, it is pertinent to run a series of tests to ensure that the process is as efficient as possible, to minimize the time invested in running them. Such tests study the resolution, number of snapshots, and abundance range to be employed. Now, for reference, the n = 00017 snapshot of the `t45g15m2005` model is run at resolution 48x48x240 for the two available isotopic ratios, that is 0% and 6%. Comparing them to one another provides us with an estimation of the effects of the isotopic ratio on the lines, which is shown in Figure 1. As expected, the isotopic ratio does not have any effect on the subordinate line since there is no isotopic splitting in it. However, for the resonance line, its effects are of the order of 10^{-1}.

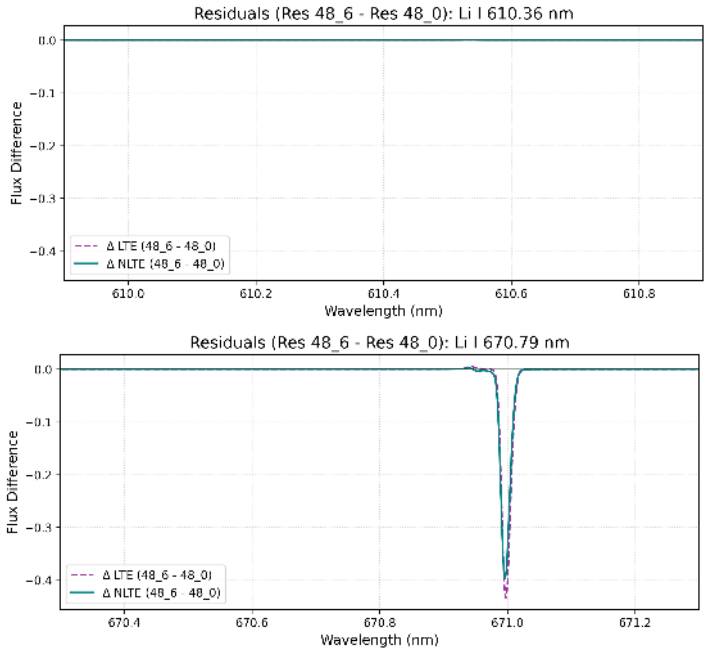


Figure 1: Residuals between isotopic ratios of 0% and 6% for the snapshot n = 00017 of the `t45g15m2005` model at resolution 48x48x240 for the subordinate line (Top) and the resonance one (Bottom). Solid teal lines represent the results for the NLTE calculations, whereas dashed purple ones represent the LTE calculations.

3.3.1. Resolution of the models

The first one of the tests performed was the resolution of the Stagger models. As already mentioned, there are two available, 48x48x240 and 80x80x240, with the latter being by definition more computationally expensive than the former one. Hence, it would be more convenient to employ 48x48x240 over 80x80x240 if the differences among them are not big compared to the effect of the isotopic ratio itself on the line of interest. Consequently, in order to test that, the `t45g15m2005` model is run for these two different resolutions and employing only one snapshot (in this case, n = 00017) using an isotopic ratio of 0%. The comparison between them is shown in Figure 2 for the two lines. As we can see, for both the subordinate and resonance line, the magnitude of the residuals between a resolution of 48x48x240 and one of 80x80x240 is of the order of 10^{-3}. So two orders of magnitude smaller than the isotopic ratio effects. Hence, it seems reasonable to employ the 48x48x240 resolution to carry out the measurements. In this sense, if not mentioned otherwise, all models calculated from now on are done so at this resolution.

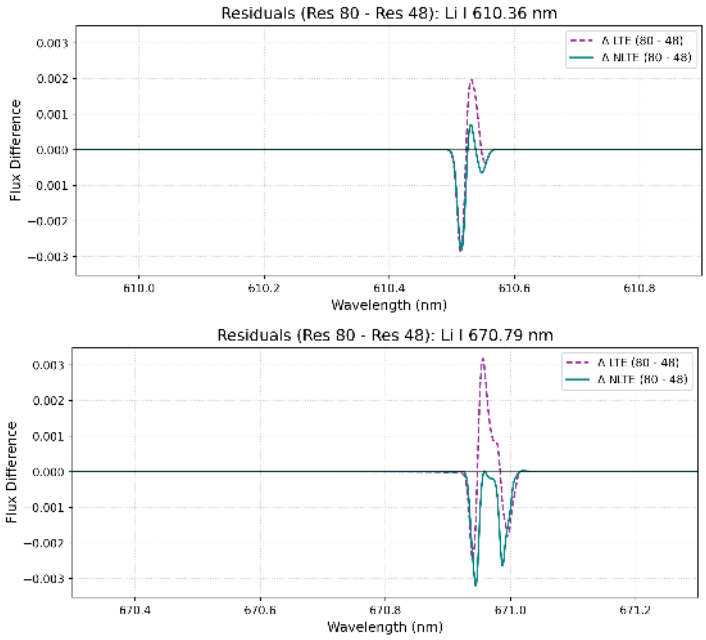


Figure 2: Residuals between resolutions for the snapshot $n = 00017$ of the `t45g15m2005` model with $\frac{^6Li}{^7Li} = 0\%$ for the subordinate line (Top) and the resonance one (Bottom). Solid teal lines represent the results for the NLTE calculations, whereas dashed purple ones represent the LTE calculations.

3.3.2. Number of snapshots

Since all the atmosphere models account for 10 different snapshots, it seems pertinent to establish an optimal number of them to use. To do that, first, a reference spectrum averaging over all of them is calculated using the `t45g15m2005` model, isotopic ratio 0%, and abundance $A(\text{Li}) = 3.42$ as derived by Sitnova et al. [2023]. Then, different spectra are calculated, averaging over 2, 3, 4, and 5 snapshots, all chosen equidistantly to sample at least one convective turnover time (Wang et al. [2022]). Note that for 5 snapshots, 2 different equidistant sets are possible, one odd and another even (i.e., taking the 1st, 3rd, 5th, 7th, and 9th snapshots or the 2nd, 4th, 6th, 8th, and 10th ones). Residuals between the reference spectra and all the averages over every set of snapshots used are shown in Figure 3 for both NLTE and LTE calculations. As we can see, residuals for the NLTE spectra are of the order of 10^{-3} for the average of 3, 4, and 5 snapshots, with the average over 4 snapshots being the one with a smaller maximum compared to the other two. In comparison to Figure 1, these are two orders of magnitude smaller, which means that they are all viable as an option for the number of snapshots to be employed. However, for this project, we decided on the average over 4 snapshots due to the residuals being slightly smaller for both NLTE and LTE.

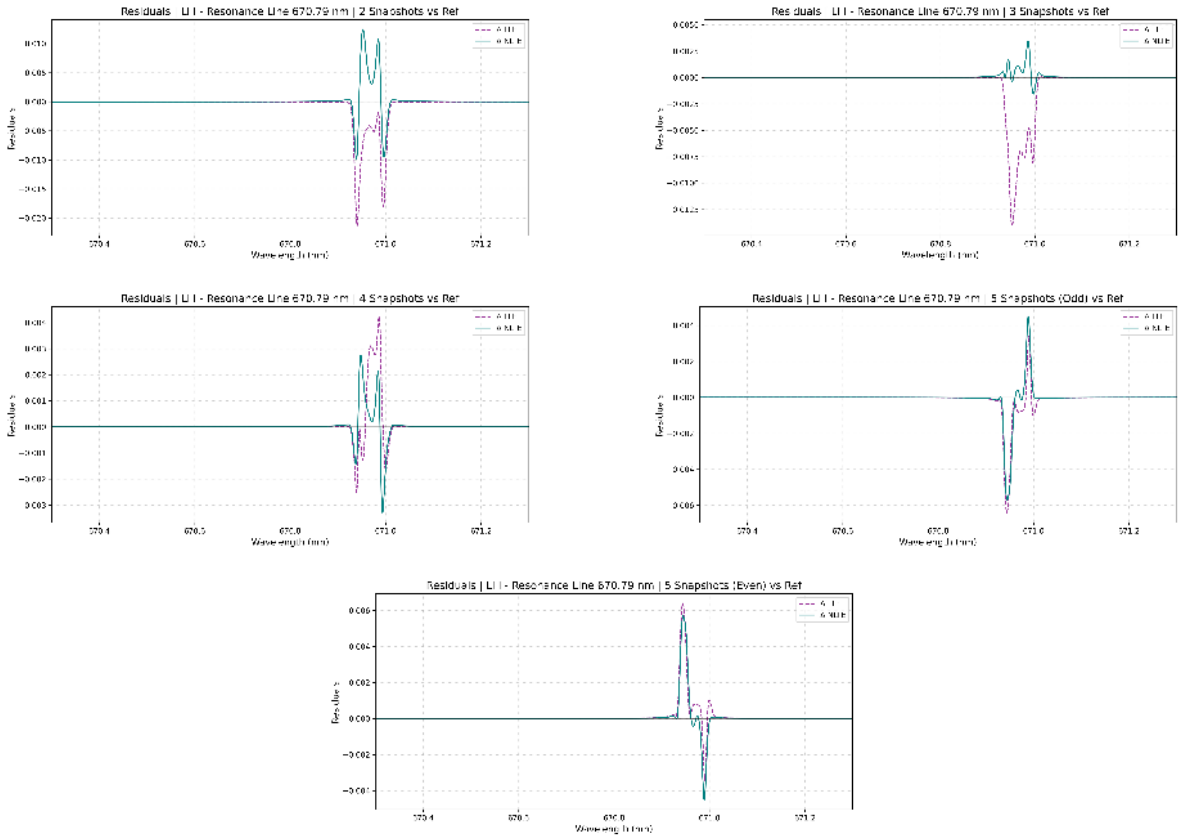


Figure 3: Residuals for the average over different snapshot numbers of the `t45g15m2005` model at isotopic ratio 0% and abundance $A(Li) = 3,42$.Top left: 2 snapshots. Top right: 3 snapshots. Middle left: 4 snapshots. Middle right: 5 snapshots (odd). Bottom: 5 snapshots (even). For all NLTE results are in teal, whereas in purple are the ones for LTE.

3.3.3. Initial abundance determination

Having set the optimal resolution and number of snapshots to be employed, it is also pertinent to set a reasonable range of abundances so that we do not do more runs than necessary. The lithium abundance as measured by Sitnova et al. [2023] is: $A(\mathrm{Li}) = 3.42 \pm 0.07$. However, having already calculated the average over 10 snapshots for the `t45g15m2005` model for this abundance in the previous test, it is possible to compare it with the observed spectrum, as shown in Figure 4.

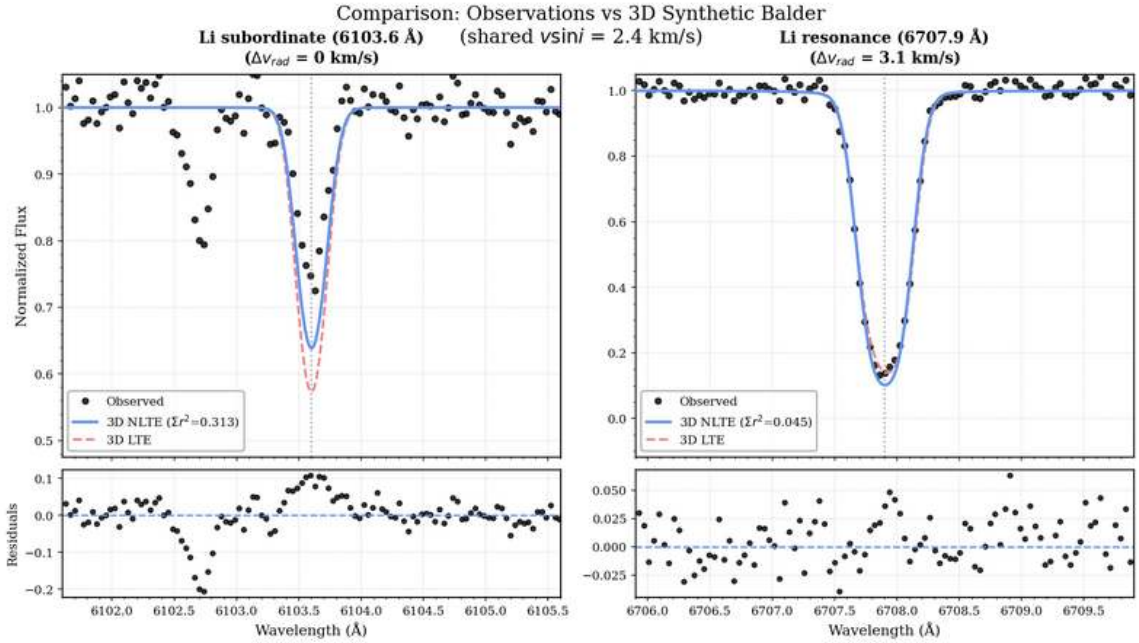


Figure 4: Observed data vs spectra of the `t45g15m2005` model averaged over 10 snapshots for $A(\mathrm{Li}) = 3.42 \pm 0.07$ and isotopic ratio of 0% for the resonance line (right) and subordinate one (left). Dashed red lines represent LTE calculations and solid blue lines NLTE ones. With Δv_{rad} and $v \sin i$ being the variation in radial velocity and the rotational broadening, respectively, which were adjusted manually to the observed spectra.

As we can see, for our models, the abundance predicted by Sitnova et al. [2023] appears to be an overestimation of such. Therefore, in order to establish this range of abundances within which our models run, it is pertinent to first give a rough initial estimate of the lithium abundance ourselves. Following Sitnova et al. [2023] such estimation is done by fitting the subordinate line since, unlike the resonance one, it does not present saturation. For that, a χ^2 fitting for the subordinate line is done using the package Breidablik (`https://github.com/ellawang44/Breidablik`), which accounts for our same grid of 3D NLTE models. The results for this are shown in Figure 5.

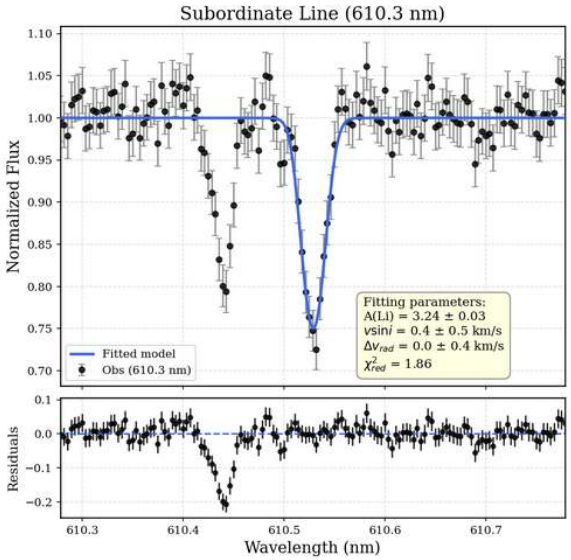


Figure 5: Breidablik, subordinate χ^2 fitting.

As observed, our initial estimate for the abundance is $A(\mathrm{Li}) = 3.24 \pm 0.03$, so smaller than Sitnova et al. [2023] by -0.18 dex. Consequently, it seems that a reasonable abundance range on which the real abundance might fall is $A(\mathrm{Li}) = [3.12, 3.42]$. Now, to perform the necessary interpolation between the spectra of the 3D models, it is pertinent to pick a small enough distance between points so that such interpolation does not incur a bigger error than the isotopic ratio effect itself. For that, the abundance points chosen have been: $A(\mathrm{Li}) = [3.12, 3.22, 3.32, 3.42]$.

3.4. Determination of the abundance and the isotopic ratio

In order to estimate the abundance and isotopic ratio, first, averaging over 4 snapshots, the spectra of the models at `t45g15m2005` and `t45g15m3005` is calculated for two isotopic ratios of 0% and 6% and four abundances $A(\mathrm{Li}) = [3.12, 3.22, 3.32, 3.42]$. This is 2 stellar models, 4 snapshots, 2 isotopic ratios, and 4 abundances, so 64 different model atmospheres, each one of them accounting for around 2 hours of computational time. As a result, we have a grid of models composed of the following points:

- $[\mathrm{Fe/H}] = [-3, -2]$
- $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = [0, 6]\%$
- $A(\mathrm{Li}) = [3.12, 3.22, 3.32, 3.42]$

Naturally, all at $T_{\mathrm{eff}} = 4500$ K and $\log g = 1.5$. Now, the measurement of the isotopic ratio is a very complicated task, since the presence of ^{6}Li on the resonance

line introduces only a very small asymmetry on the red wing of the line that is very difficult to quantify. To do it, it is then necessary to provide a simultaneous estimate of radial velocity, overall Li abundance, and rotational, convective, and instrumental broadening (Wang et al. [2022]), which results in an important degeneracy between all these parameters. To account for it and provide an estimate of the isotopic ratio, we make use of Bayesian statistics. Interpolating between the spectra from the above grid, the resulting synthetic models are fitted to the observed spectra via a Markov Chain Monte Carlo (MCMC) that provides the pertinent posterior probability function for the parameters θ.

Furthermore, the subordinate line is fitted via a similar MCMC with the same synthetic models, in order to provide a measurement for the abundance. The reason behind this is that, unlike the resonance line, the subordinate one is not saturated and lacks a measurable isotopic shift, providing a direct measurement of the total lithium abundance. Subsequently, to tackle the degeneracy between the parameters, this abundance measurement can be introduced, or not, as a prior to constrain the ^{6}Li signature in the resonance line when fitting it with the MCMC. In this way, the abundance is estimated employing the subordinate line, whereas the isotopic ratio is estimated from the resonance one in two ways, introducing the abundance as a prior and not doing so.

Additionally, the grid over which we are interpolating is truncated with respect to the $T_{\rm eff}$ and $\log g$ of the star of interest. That is $T_{\rm eff} = 4690 \pm 80$ K and $\log g = 1.34 \pm 0.13$. Recall that our only two available models between which to interpolate were `t45g15m2005` and `t45g15m3005`. To confront this issue, we employ the package Breidablik to perform a first-order linear extrapolation of our synthetic models to the ones from the star of interest. As a consequence of all this, two different MCMCs are done to fit both lines, and hence to provide estimates of the abundance and isotopic ratio. One employing the models without extrapolation, and a second one with models that do. Consequently, the details of the MCMC used are explained in the following section, and the details for the non-extrapolated and extrapolated cases are explained in Sections 3.4.2 and 3.4.3, respectively.

3.4.1. Markov Chain Monte Carlo (MCMC)

The MCMC employed for sampling in this project is the one from the Python package *emcee* (Foreman-Mackey et al. [2013]), which employs the Affine-Invariant Ensemble Sampler algorithm (Goodman and Weare [2010]). As explained, the objective of our MCMC is to sample the posterior probability function of all the free parameters θ when fitting the synthetic spectrum to the observed one of either subordinate or resonance line, that is:

$$logP(\theta|D) \propto \log P(D|\theta) + \log P(\theta) \tag{1}$$

This quantifies the probability density of the set of parameters θ given the empirical observations D. Now, Equation 1 is composed of two different functions, the likelihood one, $\log P(D|\theta)$, and the prior one, $\log P(\theta)$. Those are defined as follows:

- **Likelihood function** ($\log P(D|\theta)$): Measures the probability density of the observed data D given the parameters θ, or, in other words, how well does the model fit the observed data. It is defined according to the Gaussian logarithm:

$$\ln(L) = -0.5 \cdot \sum_{i=1}^{N} \left[\left(\frac{r_i}{\sigma} \right)^2 \right] \tag{2}$$

  With r_i being the residuals between the synthetic spectrum and the observed data, and σ being the standard deviation of the observed flux error.

- **Prior function** ($\log P(\theta)$): Is the probability distribution that represents the prior beliefs before fitting the data. Depending on the line being fitted, we employ one or two different prior functions. Firstly, in the case of the subordinate line, only a flat one, which assigns the same probability (unity) to all parameters and none to the outside ones, is used. Secondly, in the case of the resonance one, the flat prior is also used, but additionally, the abundance measured with the subordinate one $\mu_{A(\mathrm{Li})}$ is introduced as a Gaussian prior of the form:

$$\ln P(\theta_{\mathrm{Li}}) = -\frac{1}{2} \left(\frac{A(\mathrm{Li}) - \mu_{A(\mathrm{Li})}}{\sigma_{A(\mathrm{Li})}} \right)^2 \tag{3}$$

  Where $\sigma_{A(\mathrm{Li})}$ is the uncertainty of the abundance measured. The above equation assigns a lower probability to the values further from $\mu_{A(\mathrm{Li})}$ and a higher one to the ones close to it.

Now, to converge to a certain set of parameters or another, the MCMC from the *emcee* package employs, as already mentioned, the Affine-Invariant Ensemble Sampler algorithm that works as follows: For one walker, for each step, having calculated the posterior probability function for each set of parameter, the algorithm proposes a new combination of parameters and measures the difference in probability between them,

that is:

$$\Delta \ln P = \ln P(\theta_{\rm new}|D) - \ln P(\theta_{\rm old}|D) \tag{4}$$

If $\Delta \ln P > 0$, the new set of parameters is accepted. If $\Delta \ln P < 0$, however, the new set of parameters is accepted with probability equivalent to:

$$\alpha = e^{\Delta \ln P} \tag{5}$$

To implement this probabilistically, a random number u is generated from a uniform distribution spanning inside the interval $[0, 1]$. The proposed new set of parameters $\theta_{\rm new}$ is then accepted if $u \leq \alpha$; otherwise, it is rejected. In our case, for all MCMCs done, 32 walkers were employed, with 5000 steps and 1000 of burn-in time to better reach convergence.

3.4.2. Non-extrapolated case

The free parameters to be sampled are $\theta = (A(\mathrm{Li}), \frac{^6\mathrm{Li}}{^7\mathrm{Li}}, v \sin i, \Delta v_{rad})$ in the resonance line case, and $\theta = (A(\mathrm{Li}), v \sin i, \Delta v_{rad})$ in the subordinate line, since the latter is not affected by the isotopic ratio. In this sense, to quantify the likelihood function, for each step in the MCMC, that is, given a specific set of parameters, an interpolated model from the grid is extracted based on $A(\mathrm{Li})$ and $\frac{^6\mathrm{Li}}{^7\mathrm{Li}}$ and at a fixed metallicity of -2.43. Then it is broadened and shifted according to the corresponding $v \sin i$ and Δv_{rad}. This synthetic model is compared to the observed spectra, calculating its residuals and, hence, following Equation 2, its likelihood function. Subsequently, the prior function is calculated, using only a flat prior for the subordinate line, and both a flat and a Gaussian prior for the resonance one in accordance with Equation 3. Ultimately, putting the two together, following Equation 1, the posterior probability function is estimated for each line and each prior. Following this process, an estimation for the abundance is given, and two for the isotopic ratio.

3.4.3. Extrapolated case

Extrapolation of the `t45g15m2005` and `t45g15m3005` models to the parameters of our star is done as follows. First, employing, the package Breidablik, a training grid of

models between $T_{\rm eff}$ = 4000 K and $T_{\rm eff}$ = 4500 K and $\log g = 2.0$ and $\log g = 1.5$ are calculated at metallicity -2.43 and for abundances A(Li) = $[3.12, 3.22, 3.32, 3.42]$. Step size in effective temperature is 100 K, and in surface gravity is 0.1. Then, working with the depression instead of the flux values for mathematical simplicity, a matrix that stores the spectrum for each combination of $T_{\rm eff}$ and $\log g$ within the training grid is created for each abundance. Subsequently, for each pixel of the spectrum, the depression values from all the matrix models for a certain abundance are taken, and a 2D bilinear surface is created. This surface, generated for each pixel of the spectrum generated for each abundance, is then expanded according to its tendency and evaluated at $T_{\rm eff}$ = 4690 K and $\log g = 1.34$. Then, the depression values at every pixel at that $T_{\rm eff}$ and $\log g$ are evaluated against the ones at $T_{\rm eff}$ = 4500 K and $\log g = 1.5$ to generate a grid of ratios for A(Li) = $[3.12, 3.22, 3.32, 3.42]$. Secondly, this grid of ratios is introduced in the MCMC, after extracting a model interpolated from the grid, but before broadening and shifting it. Consequently, both the likelihood function and prior function are estimated in a similar way as for the non-extrapolated case, and so the posterior function for the pertinent parameters of the two lines with the corresponding priors is calculated. Therefore, again, as a result, one estimation for the abundance is obtained, and two for the isotopic ratio.

4. Results

As mentioned in Section 3.4, two MCMCs are performed in order to give an estimate of the abundance and isotopic ratio, one with the models without extrapolation and another with the extrapolated ones. The abundance is estimated by fitting the subordinate line, including only a flat prior, and its results are presented in Section 4.1. On the other hand, the isotopic ratio is estimated by fitting the resonance line, implementing a flat prior and a Gaussian one from the abundance for both cases. In Section 4.2, results for the resonance line fitting in the non-extrapolated case are presented for both priors. Similarly, results for the extrapolated case are shown in Section 4.3, and in addition, a comparative analysis is presented in Section 4.4.

4.1. Subordinate line fitting

The corner plots for the MCMC fitting of the subordinate line with both non-extrapolated and extrapolated models are shown in Figure 6.

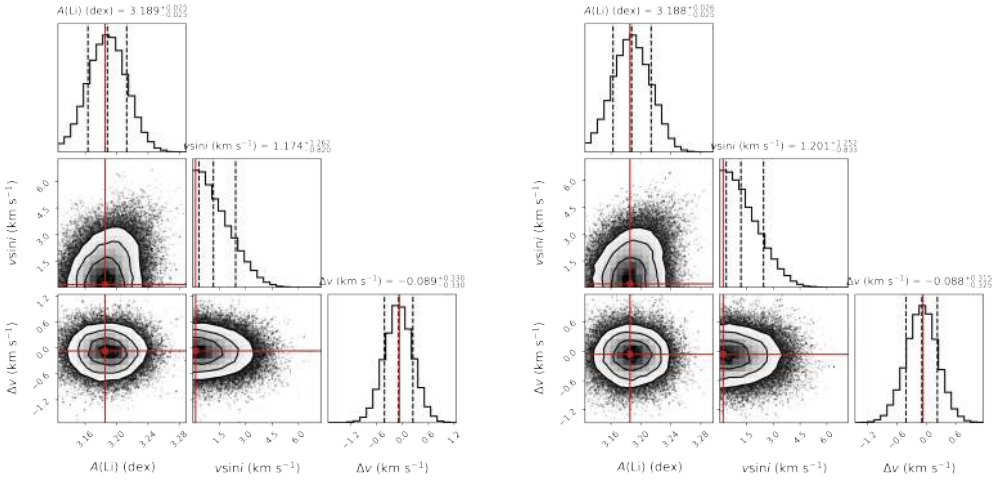


Figure 6: Corner plots for the subordinate fitting using the non-extrapolated models (Left) and extrapolated ones (Right).

Subsequently, for both of them, the results of the fittings, in conjunction with their diagnostics, are shown in Table 1.

Table 1: Results of the MCMC fitting of the subordinate line for both the non-extrapolated and extrapolated case.

	Non-Extrapolated (68% CI)		**Extrapolated (68% CI)**	
Parameter	**Median**	**Best Fit**	**Median**	**Best Fit**
A(Li) (dex)	$3.189^{+0.025}_{-0.025}$	$3.186^{+0.026}_{-0.026}$	$3.188^{+0.026}_{-0.025}$	$3.186^{+0.026}_{-0.026}$
$v \sin i$ (km s^{-1})	$1.174^{+1.262}_{-0.820}$	$0.164^{+1.326}_{-0.401}$	$1.201^{+1.251}_{-0.833}$	$0.199^{+1.712}_{-0.038}$
$\Delta V_{\rm rad}$ (km s^{-1})	$-0.089^{+0.330}_{-0.330}$	$-0.071^{+0.331}_{-0.331}$	$-0.088^{+0.315}_{-0.325}$	$-0.067^{+0.328}_{-0.328}$
Diagnostics				
χ^2	5.839		5.842	
Acceptance Rate	0.615		0.617	
Independent Samples	~ 89		~ 89	

As observed in both corner plots, the ellipses are not very eccentric and have zero slope; hence, the degeneracy between the parameters is negligible. Additionally, the acceptance rate and autocorrelation time are very good, indicating that the MCMC chains mixed very well and explored the parameter space effectively. However, both χ^2 values are far from 1.0; this suggests that, although the MCMC found the best possible minimum, the model does not exactly fit all the observed data. A possible explanation for this might be an underestimation of the errors since the only uncertainty introduced in the MCMC is the one from the observed flux. Finally, the spectral fitting for both cases is shown in Appendix 1 along with plots for the residuals of this fitting and trace plots of the MCMC.

4.2. Resonance line fitting: non-extrapolated case

The corner plots for the MCMC fitting of the resonance line with non-extrapolated models are shown in Figure 7 for both priors:

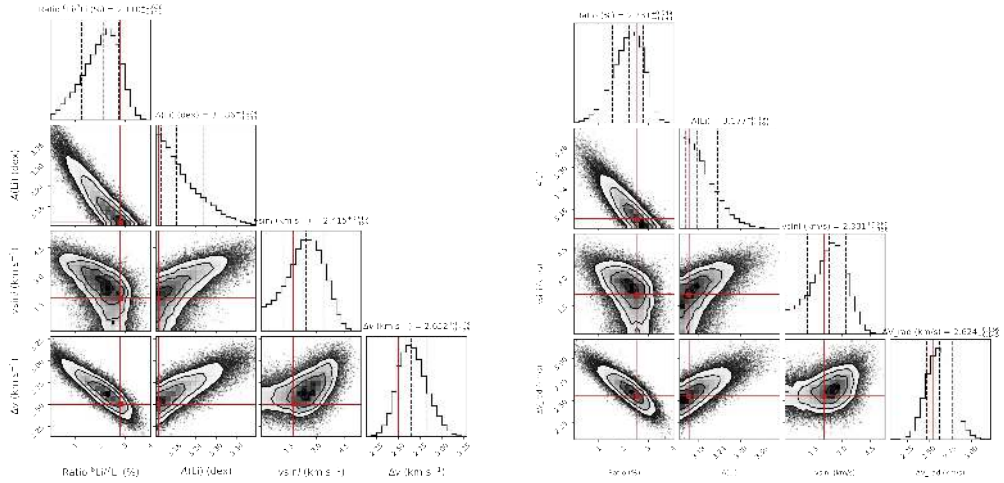


Figure 7: Corner plots for the resonance fitting with non-extrapolated models for flat prior (left) and Gaussian one (right).

Subsequently, the results of the fittings, in conjunction with their diagnostics, is shown in Table 2.

Table 2: Results of the MCMC fitting of the resonance line with non-extrapolated models for both a flat and a Gaussian prior.

	Flat Prior (68% CI)		**Gaussian Prior (68% CI)**	
Parameter	**Median**	**Best Fit**	**Median**	**Best Fit**
^{6}Li/^{7}Li (%)	$2.110^{+0.636}_{-0.850}$	$2.803^{+1.127}_{-1.154}$	$2.231^{+0.548}_{-0.680}$	$2.555^{+0.614}_{-0.614}$
A(Li) (dex)	$3.186^{+0.079}_{-0.044}$	$3.134^{+0.090}_{-0.002}$	$3.177^{+0.057}_{-0.036}$	$3.152^{+0.044}_{-0.044}$
$v \sin i$ (km s^{-1})	$2.415^{+0.962}_{-1.185}$	$1.772^{+1.333}_{-1.333}$	$2.331^{+0.860}_{-1.158}$	$2.065^{+0.994}_{-0.994}$
$\Delta V_{\rm rad}$ (km s^{-1})	$2.652^{+0.178}_{-0.155}$	$2.504^{+0.235}_{-0.235}$	$2.624^{+0.156}_{-0.145}$	$2.556^{+0.150}_{-0.150}$
Diagnostics				
χ^2	1.188		1.190	
Acceptance Rate	0.538		0.546	
Independent Samples	~ 62		~ 68	
Lithium Detection Significance (^{6}Li/^{7}Li)				
95.45% confidence interval ($\sim 2\sigma$)	[0.445, 3.216] %		[0.821, 3.226] %	

Additionally, the plot for the best fit is shown in Figure 8.

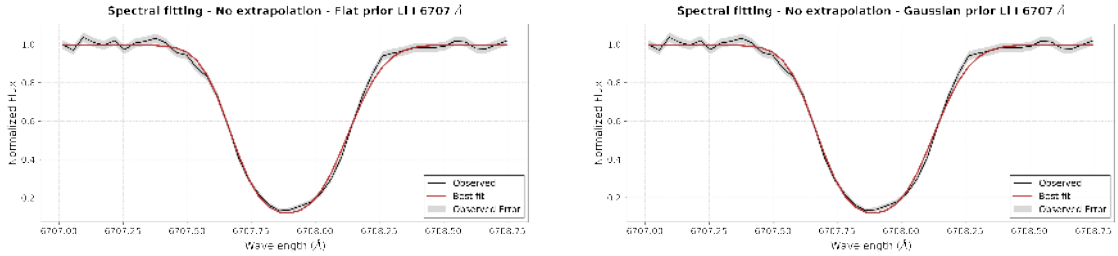


Figure 8: Best fit result for the resonance line with non-extrapolated models for a flat prior (left) and a Gaussian one (right).

Both corner plots show a very strong anti-correlation between the isotopic ratio and abundance, with a pronounced negative slope and a very eccentric ellipse, meaning both parameters are strongly degenerated. In this sense, the rest of the parameters, even though in a more mild way, are still either correlated or anti-correlated with one another, showing steep slopes and eccentric ellipses in all cases. One notable case is that of $v \sin i$, which seems to be nonlinearly related to both abundance and ratio. In addition, in both cases, the χ^2 diagnostic yields a slightly underfitting $\chi^2 \sim 1.188$, with similar autocorrelation times for all parameters and a slightly high acceptance rate. Meaning that the chains mixed well with reasonable step sizes. Finally, residuals for the spectral fitting as well as trace plots for this case are presented in Appendix 2.

4.3. Resonance line fitting: extrapolated case

The corner plots for the MCMC fitting of the resonance line with extrapolated models are shown in Figure 9.

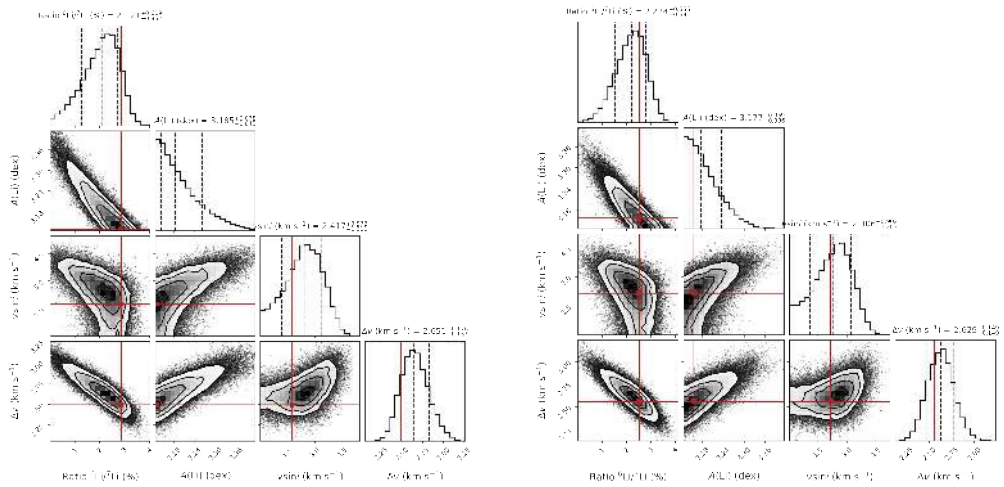


Figure 9: Corner plot for the resonance fitting with extrapolated models for flat prior (left) and Gaussian one (right).

Subsequently, the results of the fittings, in conjunction with their diagnostics, is shown in Table 3.

Table 3: Results of the MCMC fitting of the resonance line with extrapolated models for both a flat and a Gaussian prior.

	Flat Prior (68% CI)		**Gaussian Prior (68% CI)**	
Parameter	**Median**	**Best Fit**	**Median**	**Best Fit**
^{6}Li/^{7}Li (%)	$2.121^{+0.620}_{-0.861}$	$2.858^{+0.691}_{-0.691}$	$2.224^{+0.556}_{-0.681}$	$2.519^{+0.619}_{-0.619}$
A(Li) (dex)	$3.185^{+0.078}_{-0.043}$	$3.129^{+0.050}_{-0.050}$	$3.177^{+0.058}_{-0.036}$	$3.155^{+0.044}_{-0.044}$
$v \sin i$ (km s^{-1})	$2.417^{+0.972}_{-1.211}$	$1.749^{+1.066}_{-1.066}$	$2.306^{+0.902}_{-1.203}$	$2.164^{+1.024}_{-1.024}$
$\Delta V_{\rm rad}$ (km s^{-1})	$2.651^{+0.176}_{-0.157}$	$2.492^{+0.163}_{-0.163}$	$2.629^{+0.149}_{-0.149}$	$2.560^{+0.149}_{-0.149}$
Diagnostics				
χ^2	1.188		1.191	
Acceptance Rate	0.535		0.545	
Independent Samples	~ 59		~ 62	
Lithium Detection Significance (^{6}Li/^{7}Li)				
95.45% confidence interval ($\sim 2\sigma$)	[0.404, 3.210] %		[0.824, 3.261] %	

Additionally, the plot for the best fit is shown in Figure 10.

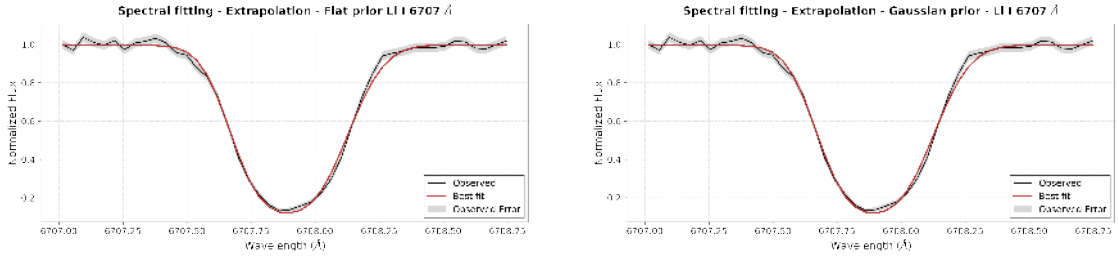


Figure 10: Best fit results for the resonance line with extrapolated models for a flat prior (left) and a Gaussian one (right).

The results of the corner plots, as well as the discussion of the diagnostics, are very similar as in the non-extrapolated case. Similarly to before, residuals for the best fit as well as the trace plots for the MCMC are shown in Appendix 3.

4.4. Comparative Analysis

When fitting the resonance line, the results for the best fit of the isotopic ratio are very similar between the non-extrapolated and extrapolated cases. For both of them, the values obtained using the flat prior are virtually the same, and in the two cases, lower than the Gaussian prior ones, which are also very similar to each other. Hence, extrapolation does not seem to have a significant impact when estimating the isotopic ratio. The priors, however, do impact such estimation, with the Gaussian one lowering it by roughly 0.3 percentage points in both cases. Additionally, the estimations of the abundance for the non-extrapolated and extrapolated cases when fitting the subordinate line are equal.

Moreover, it is also true that for both the best fit of $v \sin i$ and Δv_{rad}, the differences between the extrapolated and non-extrapolated cases and the priors are minimal in the resonance line. The differences when comparing them to their respective best fits from the subordinate line are considerable. Firstly, for $v \sin i$, the resonance line is heavily saturated, contrary to the subordinate one. Adding lithium to a saturated line will not deepen it, but rather, broaden it, which is the same effect that $v \sin i$ has on the line. Hence, leading to a strong degeneracy between the parameters. This can be seen in the corner plots of Figures 6, 7, and 9 since in the case of the resonance line, there is a nonlinear correlation between abundance and $v \sin i$ that is not present in the subordinate line. Additionally, the resonance line accounts for ^{6}Li, which is not present in the subordinate line. In this sense, as seen in the corner plots, the correlation between $v \sin i$ and $\frac{^6\text{Li}}{^7\text{Li}}$ is also quite complex. The ^{6}Li signature of a small excess absorption in this same saturated, broadened region partially competes with $v \sin i$ for the same feature, deepening the degeneracy.

Furthermore, for Δv_{rad}, given the correlation between the parameters for the resonance line as seen in Figures 7 and 9 and also, similarly to the case of $v \sin i$, given the introduction of ^{6}Li, its estimation in both lines is also different. On top of this, in both the non-extrapolated and extrapolated cases, the full ellipse of the abundance is not fully sampled for the resonance line, which alters the estimation of both $v \sin i$ and Δv_{rad} given the correlation of the abundance with both of them. Moreover, for both $v \sin i$ and Δv_{rad}, it is important to consider two factors that also have an impact on the estimations. First, the S/N between the lines is different and considerably low in both cases, and secondly, the calibration between the two lines is different given their spectral separation. Additionally, in all MCMCs, the best fit seems to be overestimating the absorption, especially in the red wing of the line, where the isotopic feature lies. To compensate for this, $v \sin i$ should be lowered in conjunction with the isotopic ratio to better adjust to the observed data. Finally, as shown in the Appendices, the trace plots in all cases seem to appropriately converge before the burn-in ends for all walkers.

5. Discussion

In stars with a very high lithium content ($A(\mathrm{Li}) > 3.0$), such as our case, the Li I resonance doublet at 6707 Å becomes highly saturated, meaning it falls on the flat part of the curve of growth where its equivalent width is inherently insensitive to abundance changes. This insensibility increases the error when measuring $A(\mathrm{Li})$ and also worsens the degeneracy between the abundance and the broadening parameters $v \sin i$ as both compete for the same effect on the line width once the core is saturated. This degeneracy is extended to the isotopic ratio, given that the signature of ^{6}Li is a small absorption excess on the red wing of the line, and also to Δv_{rad} due to it compensating that signature. This degeneracy can be visualized in all corner plots of Sections 4.2 and 4.3, and is non-existent in the ones shown in Section 4.1 for the much weaker subordinate line at 6103 Å given the lack of saturation and isotopic splitting in this line. Subsequently, in order to tackle this degeneracy, the abundance is estimated by fitting this subordinate line and then employed as a prior to constrain the abundance when fitting the resonance one. Note, however, that this prior does not tackle the natural partial degeneracy between $\frac{^6\mathrm{Li}}{^7\mathrm{Li}}$, $v \sin i$, and Δv_{rad} .

In this sense, two different MCMCs were employed to fit the subordinate line and hence derive the abundance, one employing non-extrapolated models and another one using extrapolated ones. The result for the best fits for the abundance was for both $A(\mathrm{Li}) = 3.186 \pm 0.026$. This estimation accounts for a difference with respect to the value predicted by Sitnova et al. [2023], which is 3D corrected, of -0.234 dex. Note that, on one side, we are fitting with a model that is cooler and has higher surface gravity than the star of reference, and on the other, we are fitting with extrapolated models. Surface gravity difference is small, and so are its effects on the line; however, temperature effects are more significant. Unfortunately, there are no estimates in the literature of the error induced in the measurement of the abundance by differences in temperature employing 3D NLTE models in VMP Li-rich giants. On their side, Sitnova et al. [2023] measured this error when fitting for the subordinate line with 1D NLTE models corrected by 3D effects. The value found by them is of ~ 0.05 for a difference of $\Delta T_{\mathrm{eff}} = \pm 80$K and therefore of ~ 0.118 for one of $\Delta T_{\mathrm{eff}} = 190$ K, such as ours. On our side, employing the package Breidablik, this error was estimated directly for a difference of $\Delta T_{\mathrm{eff}} = 190$ K and was found to be ~ 0.17. Hence, it seems likely that our ΔT_{eff} should not induce a difference of -0.234 dex as the one between our estimation and Sitnova et al. [2023] when measuring the abundance from the subordinate line with 3D NLTE models in our star. Furthermore, in what regards to our first order linear extrapolation, errors in the abundance estimation come from the non-linearity excess of the response to T_{eff} and $\log g$ of our line. Multiplying a 3D spectrum by a grid of scalar

ratios assumes that the spectral line reacts linearly; however, 3D convection and NLTE effects are highly nonlinear. Still, for a gap of the size of ours in $T_{\rm eff}$ and $\log g$, such a response can be assumed to remain highly linear, so, in any case, the extrapolation error should not account for a -0.234 dex difference.

As a consequence of all this, the difference between Sitnova et al. [2023] estimation and ours is likely to arise from methodological reasons. For their estimation, Sitnova et al. [2023] derived the abundance through a classical 1D NLTE profile-fitting approach and applied, subsequently, analytical 3D corrections from Breidablik (Wang et al. [2022]). The treatment of the radiative transfer between 1D NLTE and 3D NLTE is fundamentally different; 1D model atmospheres employ MLT and assume hydrostatic equilibrium, not accounting for time evolution. On the other hand, 3D model atmospheres, such as the ones employed in this project, solve the radiative hydrodynamics equations (RHD) in a three-dimensional simulation box over time. Additionally, for 1D NLTE models, two artificial parameters must be introduced to account for the Doppler broadening of the line due to convection, macroturbulence v_{mac}, and microturbulence ξ. This second parameter is known to de-saturate the lines, changing their equivalent width (EW) and directly altering the derived abundance, and must be calibrated manually with the corresponding uncertainty. On the other hand, 3D models do not need the implementation of these parameters since Doppler broadening is naturally calculated by the code itself. Furthermore, the 3D corrections introduced by Sitnova et al. [2023] consist of a scalar Δ_{3D-1D} that does not take into account the topology of the line, with their abundance estimation remaining dependent on the two manually introduced parameters ξ and v_{mac} . Lastly, the difference in the abundance estimation between Sitnova et al. [2023] and ours may arise from the fitting procedure followed. In their study, Sitnova et al. [2023] employed a fragmented χ^2 minimization, whereas in this project, we employed an MCMC. Both methods do account for covariance, but χ^2 searches for the local minima, whereas MCMC explores the parameter space, accounting for the degeneracy between the parameters. Still, in the case of the best fit, they should give the same result.

On another note, after estimating the abundance with the subordinate line, its value is used as a Gaussian prior in the MCMC fitting of the resonance line, in addition to the flat one, to constrain the isotopic ratio signature. Note that no other prior of this kind is used, as could be the one for $v \sin i$, given the difference in calibration between the lines. For the fitting with the non-extrapolated models, the best fit for the isotopic ratio found when employing the flat prior was $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = 2.803^{+1.127}_{-1.154}\%$ whereas the one found employing the already mentioned Gaussian prior of $A(\mathrm{Li}) = 3.186$ was $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = 2.555\%^{+0.614}_{-0.614}$. This is a difference of 0.253 percentage points between them and one of 2.847 and 3.1 percentage points, respectively, between these two estimations and the estimation of Sitnova et al. [2023], that is $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = 5.65^{+5.05}_{-2.51}\%$. Moreover, for the extrapolated case, the

best fit employing the flat prior was $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = 2.858^{+0.691}_{-0.691}\%$ and employing the Gaussian prior was $\frac{^6\mathrm{Li}}{^7\mathrm{Li}} = 2.519^{+0.619}_{-0.619}\%$. This is a difference of 0.338 percentage points between them and one of 2.792 and 3.131 percentage points, respectively, with Sitnova et al. [2023]. Now, recalling the histograms for the isotopic ratio in all corner plots of the resonance line, and also Tables 2 and 3, it is possible to see that all these cases are consistent with a detection of ^{6}Li with a confidence level of at least 2σ. Subsequently, all these best-fit estimations for the isotopic ratio are in contrast with the best fit provided by Sitnova et al. [2023] and lower their value significantly. Similarly, as for the case of the abundance, neither the differences introduced by fitting with a cooler model nor the ones that arise from using an extrapolated one can account for this stark difference. It is true, however, that since the package Breidablik does not account for the isotopic ratios, the extrapolation done assumes that ^{6}Li behaves exactly the same as ^{7}Li with T_{eff} and $\log g$, which is physically not true. Still, comparing the extrapolated to the non-extrapolated best fits for the isotopic ratio yields a very small difference; thus, these additional errors should not have a big impact.

For that reason, again, the disparity between our estimate and Sitnova et al. [2023] is likely to rest on the methodological reasons already explained. In this sense, for the estimation of the isotopic ratio, the impossibility for 1D models to naturally account for convection is especially important. Measuring the isotopic ratio rests on characterizing the small asymmetry on the red wing of the resonance line that the presence of ^{6}Li introduces. Such asymmetry is also produced by convection, therefore mimicking the isotopic ratio signature. MLT theory accounts only for a symmetrical broadening of the line, whereas 3D convective effects can broaden it in an asymmetrical way. For this reason, 1D NLTE models inherently misidentify this asymmetry with the presence of ^{6}Li and so, measuring the isotopic ratio with this kind of model yields an unreliable result even when correcting *a posteriori* for 3D effects. In relation to this, the scalar correction applied by Sitnova et al. [2023] shifts the mean best-fit abundance of the resonance line upward but leaves the 1D profile shape unchanged. The convective red-wing asymmetry, therefore, remains unmodeled, and the resulting abundance discrepancy between the two lines is interpreted entirely as ^{6}Li. Moreover, the Breidablik package employed by Sitnova et al. [2023] to implement these 3D corrections does not account for the presence of ^{6}Li and assumes that the impact of the isotope on the corrections is small compared to the corrections themselves, which is physically not true. Critical of their assumption, Sitnova et al. [2023] admit that a 3D NLTE analysis, such as the one done here, is needed before we can confirm the presence of ^{6}Li in the star, warning that their method has never been tested on Li rich stars.

Having performed this self-consistent 3D NLTE analysis, we do detect ^{6}Li with at least a 2σ confidence level, but in a significantly smaller quantity than Sitnova et al. [2023] does. As mentioned in the introduction, in a VMP giant, the first dredge-up and

RGB deplete Li. The fact that the star is rich in ^{7}Li is consistent with a scenario of recent production; however, the presence of ^{6}Li would be inexplicable due to two main reasons. First, there is no known ^{6}Li production channel in these kinds of stars, and secondly, even if there was such a channel, the first dredge-up would have destroyed it. It is true that lithium could have been produced or gained after this dredge-up; hypothetically, this could have happened by in situ spallation or exoplanet engulfment. On one hand, the in situ spallation should be negligible in an old VMP giant since it is supposed to decrease with age (Tatischeff and Thibaud [2007]); however, following the discussion from Sitnova et al. [2023], this could be checked by obtaining an additional observed spectrum and identifying a variation, or not, in the resonance line. On the other hand, stars and their planetary systems form coevally from the same parental molecular cloud, meaning they share an identical initial chemical inventory, which should be nearly free of ^{6}Li according to BBN predictions. So, pollution by planet engulfment could only be possible as an explanation if an external planet with sufficiently high ^{6}Li content was engulfed, such as a rogue brown dwarf. Nevertheless, this hypothesis, while possible, is highly unlikely. As a consequence of all this, detecting ^{6}Li on the VMP Li-rich giant Gaia DR3 1918529631627603072 would imply that either the current stellar evolution models are erroneous, since the first dredge-up did not deplete ^{6}Li. Or that something is missing in nucleosynthetic theory, given that after the first dredge-up, a new unknown production channel generated the ^{6}Li.

6. Conclusions

The aim of this project was to reanalyze the anomalous detection of ^{6}Li in Gaia DR3 1918529631627603072 from Sitnova et al. [2023], which, employing 1D NLTE models with 3D corrections from Breidablik, reported an isotopic ratio of $\frac{^6\text{Li}}{^7\text{Li}} = 5.65^{+5.05\%}_{-2.51\%}$. In our case, we employed self consistent 3D NLTE models, using the 3D Stagger grid in conjunction with the NLTE code Balder, and, given the lack of grid models close to the specific stellar parameters ($T_{\text{eff}} = 4690 \pm 80$ K, $\log g = 1.34 \pm 0,13$ and $[\text{Fe/H}] = -2.43 \pm 0,07$) as to interpolate, did the analysis for two different cases. One, using models at a cooler T_{eff} and lower $\log g$, and another one, employing extrapolation for such models to the parameters of the star of interest. In both of them, the analysis was carried out via a Markov Chain Monte Carlo (MCMC) fitting of the resonance line, taking into account two different priors: a flat one and a Gaussian one for the abundance. This prior was introduced to tackle the degeneracy, having discarded another potential similar prior as could be the one for $v \sin i$ given the difference in the calibration of the lines. As a result, we report a detection of ^{6}Li with at least a 2σ confidence level in all cases, with a fiducial value for the best fit of the isotopic ratio of $\frac{^6\text{Li}}{^7\text{Li}} = 2.519^{+0.619}_{-0.619}$ for the fitting done with the extrapolated models and the Gaussian prior. The reason for this choice is its proximity to the pertinent stellar parameters as well as the reduction in degeneracy that the constraint in abundance implies. This value is lower than the one reported by Sitnova et al. [2023] of $\frac{^6\text{Li}}{^7\text{Li}} = 5.65^{+5.05}_{-2.51}\%$ by 3.131 percentage points. Additionally, the abundance was measured by fitting the subordinate line with a similar MCMC and was found to be, for both the non-extrapolated and extrapolated cases, $A(\text{Li}) = 3.186 \pm 0.026$, so also lower than the estimate provided by Sitnova et al. [2023] of $A(\text{Li}) = 3.42 \pm 0.07$ by, in this case, 0.234 dex.

These differences are not likely to come from the errors that arise from either the truncation of the grid of models employed to fit, nor the extrapolation ones, but rather, they seem to arise from methodological reasons. In relation to this, 1D models cannot naturally account for convection, which mimics the ^{6}Li signature on the resonance line, and the 3D corrections employed by Sitnova et al. [2023] do not resolve this. So, when using 1D models for the measurements, the 3D convective effects are misinterpreted as ^{6}Li, leading to an overestimation of the isotopic ratio. That is why, when measuring $\frac{^6\text{Li}}{^7\text{Li}}$, 3D NLTE models are inherently more accurate than the 1D NLTE ones, even when corrected. Still, our methodology suffers from limitations regarding grid size and also extrapolation that must be addressed in order to confirm our reported detection. On one hand, running Stagger to create new grid points over which to perform interpolation to the specific stellar parameters of the star would be of importance; however, ideally, it would be pertinent to compute a specific 3D NLTE model at $T_{\text{eff}} = 4690$ K and

$\log g = 1.34$. On the other hand, were this not possible, our extrapolation could be improved, since we use Breidablik for it with the inherent assumption of ^{6}Li behaving as ^{7}Li, which is not true. To palliate that, in the future, it would be of interest to run the models between $T_{\rm eff} = [4000, 4500]$ K and $\log g = [2.0, 1.5]$ that do exist in the Stagger grid, accounting for the presence of ^{6}Li, and then follow a similar extrapolation methodology as in Section 3.4.2. Additionally, in future work, it would also be of interest to account for the systematic errors via a nuisance parameter in the MCMC since they are likely to be underestimated. Finally, in the case of the resonance line, the full ellipse for the abundance is not accounted for due to the limits of our abundance grid $A(\mathrm{Li}) = [3.12, 3.22, 3.32, 3.42]$. Subsequently, given the acute correlation between the abundance and the isotopic ratio, it would be of importance to perform more runs at a lower lithium abundance and account for the full ellipse. Consequently, before all this is done, even having employed 3D NLTE models to reanalyze the star as required by Sitnova et al. [2023], the presence of ^{6}Li in Gaia DR3 1918529631627603072 remains to be confirmed.

Acknowledgments

I would like to thank both my supervisors, Karin Lind and Ella Xi Wang, for their guidance throughout this project. Also, I would like to thank Terese Thidemann Hansen for reducing the observational data used in this work, as well as Anish Mayur Amarsi for providing access to his computational code, Balder. Lastly, this work has made use of the SMOKA science data archive, which is operated by the Astronomy Data Center, National Astronomical Observatory of Japan.

Appendices

Appendix 1: Extra plots for the subordinate line fitting

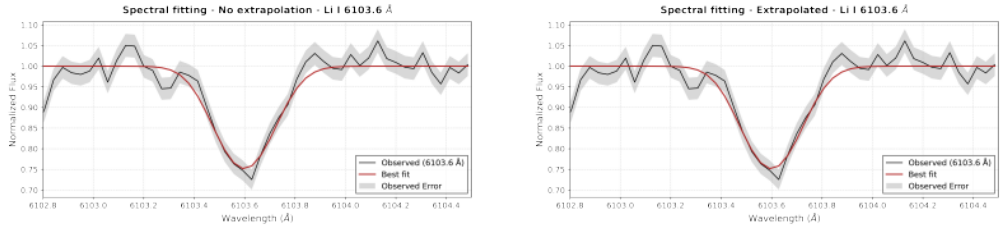


Figure 11: Best fit results for the subordinate line with non-extrapolated models (left) and extrapolated ones (right).

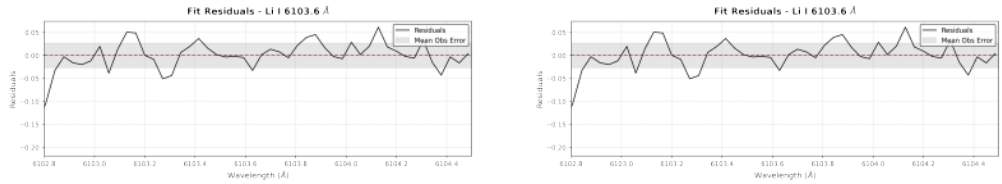


Figure 12: Residuals plot for the best fit of the subordinate line for the non-extrapolated case (left) and extrapolated one (right).

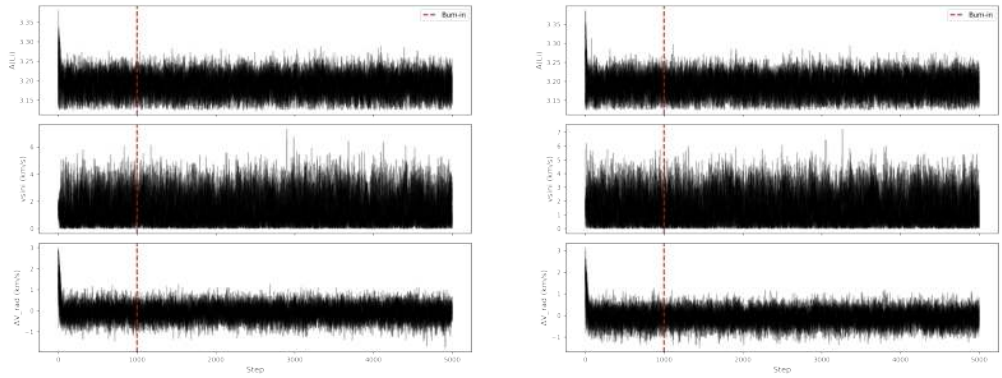


Figure 13: Trace plots for the MCMC fitting of the subordinate line with 32 walkers in the non-extrapolated case (left) and extrapolated one (right).

Appendix 2: Extra plots for the resonance line fitting with non-extrapolated models.

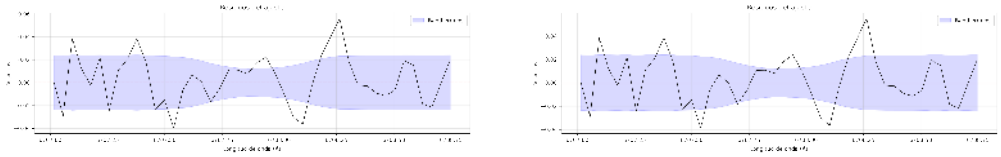


Figure 14: Residuals plot for the best fit of the resonance line in the non-extrapolated case for a flat prior (left) and a Gaussian one (right).

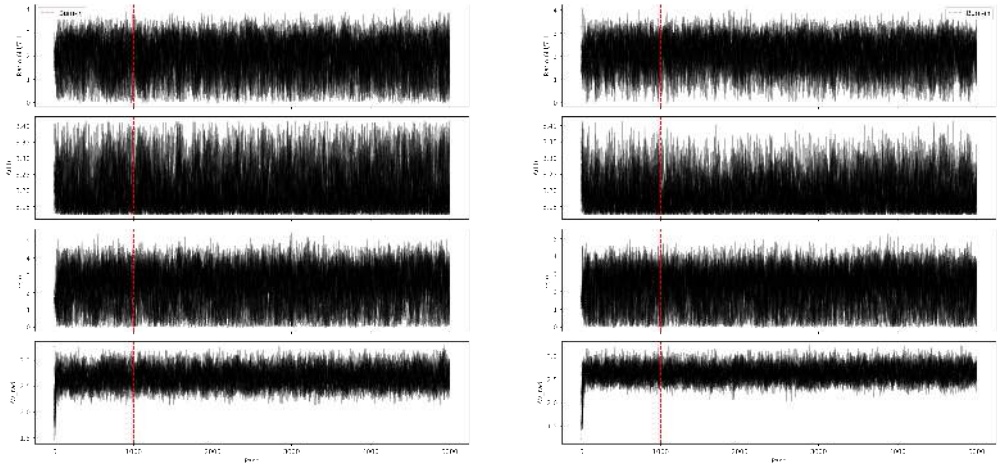


Figure 15: Trace plots for the MCMC fitting of the resonance line with 32 walkers in the non-extrapolated case for a flat prior (left) and a Gaussian one (right).

Appendix 3: Extra plots for the resonance line fitting with extrapolated models.

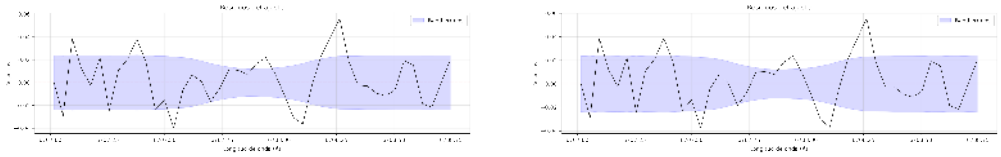


Figure 16: Residuals plot for the best fit of the resonance line in the extrapolated case for a flat prior (left) and a Gaussian one (right).

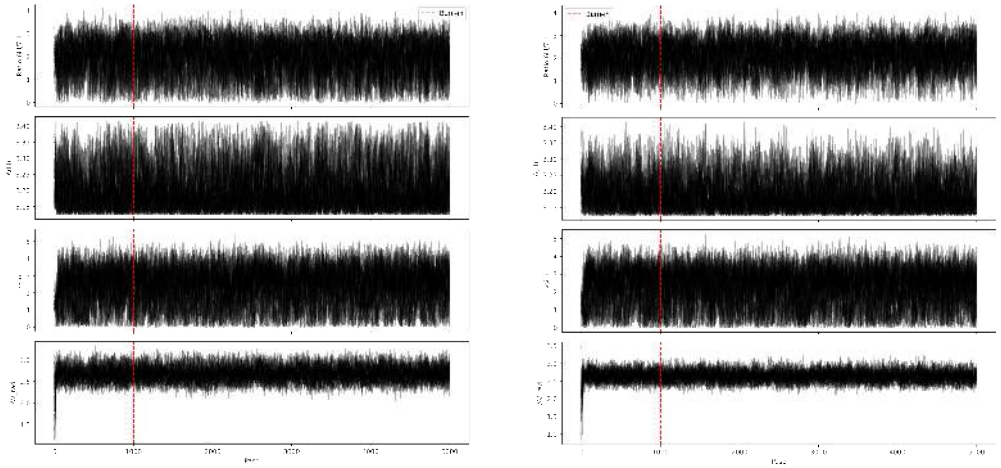


Figure 17: Trace plots for the MCMC fitting of the resonance line with 32 walkers in the extrapolated case for a flat prior (left) and a Gaussian one (right).

References

A. M. Amarsi, M. Asplund, R. Collet, and J. Leenaarts. Three-dimensional Non-LTE radiative transfer calculations of iron lines. *Monthly Notices of the Royal Astronomical Society*, 455:3735, 2016a.

A. M. Amarsi, K. Lind, M. Asplund, P. S. Barklem, and R. Collet. Non-LTE line formation of iron in late-type stars. *Monthly Notices of the Royal Astronomical Society*, 463:1518, 2016b.

A. M. Amarsi, T. Nordlander, P. S. Barklem, M. Asplund, R. Collet, and K. Lind. The Galactic evolution of oxygen based on 3D non-LTE calculations. *Astronomy and Astrophysics*, 615:A139, 2018.

M. Asplund, D. L. Lambert, P. E. Nissen, F. Primas, and V. V. Smith. Lithium Isotopic Abundances in Metal-poor Halo Stars. *The Astrophysical Journal*, 644:229, 2006.

A. G. W. Cameron and W. A. Fowler. Lithium Production in Red Giant Stars. *The Astrophysical Journal*, 164:111, 1971.

R. Cayrel et al. The lithium isotopic ratio in old halo stars and the 3D convective asymmetry. *Astronomy and Astrophysics*, 473:L37, 2007.

D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman. emcee: The MCMC Hammer. *Publications of the Astronomical Society of the Pacific*, 125:306, 2013.

X. Fu, A. Bressan, P. Molaro, and P. Marigo. Lithium depletion in metal-poor stars. *Monthly Notices of the Royal Astronomical Society*, 452:3256, 2015.

J. I. González Hernádez, P. Bonifacio, E. Caffau, H. G. Ludwig, M. Steffen, L. Monaco, and R. Cayrel. The lithium isotopic ratio in old turn-off stars. *Astronomy and Astrophysics*, 628:A111, 2019.

Jonathan Goodman and Jonathan Weare. Ensemble samplers with affine invariance. *Communications in Applied Mathematics and Computational Science*, 5:65–80, 2010. doi: 10.2140/camcos.2010.5.65.

A. J. Korn, F. Grundahl, O. Richard, P. S. Barklem, L. Mashonkina, R. Collet, n. Piskunov, and B. Gustafsson. Atomic diffusion as the solver of the cosmological lithium problem. *Nature*, 442:657, 2006.

K. Lind, J. Meléndez, M. Asplund, R. Collet, and Z. Magic. The lithium isotopic ratio in old halo stars. *Astronomy and Astrophysics*, 554:A96, 2013.

Z. Magic, R. Collet, M. Asplund, R. Trampedach, W. Hayek, A. Chiavassa, R. F. Stein, and Å. Nordlund. The Stagger-grid: A grid of 3D stellar atmosphere models. I. Methods and general properties. *A&A*, 557:A26, September 2013. doi: 10.1051/0004-6361/201321407.

A. Mott, M. Steffen, E. Caffau, F. Spada, and K. G. Strassmeier. Lithium abundance and Li/7Li ratio in the active giant HD 123351. I. A comparative analysis of 3D and 1D NLTE line-profile fits. *Astronomy and Astrophysics*, 604:A44, 2017. doi: 10.1051/0004-6361/201730409.

A. Mucciarelli, L. Monaco, P. Bonifacio, M. Salaris, X. Fu, and S. Villanova. Discovery of two Li-rich giant stars in Omega Centauri. *Astronomy and Astrophysics*, 623:A55, 2019.

n. Prantzos. Production and evolution of light elements in the Galaxy. *Astronomy and Astrophysics*, 542:A67, 2012.

O. Richard, n. Michaud, and J. Richer. Stellar models with atomic diffusion and the lithium plateau. *The Astrophysical Journal*, 619:538, 2005.

L. Sbordone et al. The lithium plateau in very metal-poor stars. *Astronomy and Astrophysics*, 522:A26, 2010.

T. M. Sitnova, T. Matsuno, Z. Yuan, N. F. Martin, P. Banerjee, F. Sestito, K. A. Venn, and J. I. González Hernández. The Pristine survey - XXII. A serendipitous discovery of an extremely Li-rich very metal-poor giant and a new method of Li/Li isotope measurement. *Monthly Notices of the Royal Astronomical Society*, 000(1):1–14, 2023.

V. V. Smith, D. L. Lambert, and P. E. Nissen. The 6Li/7Li ratio in old halo stars. *The Astrophysical Journal*, 408:262, 1993.

F. Spite and M. Spite. Abundance of lithium in unevolved halo stars and old disk stars. *Astronomy and Astrophysics*, 115:357, 1982.

V. Tatischeff and J.-P. Thibaud. Is ^{6}Li in metal-poor halo stars produced in situ by solar-like flares? *Astronomy and Astrophysics*, 469(1):265–274, 2007. doi: 10.1051/0004-6361:20066925.

E. X. Wang, T. Nordlander, M. Asplund, K. Lind, Y. Zhou, and H. Reggiani. Non-detection of ^{6}Li in Spite plateau stars with ESPRESSO. *Monthly Notices of the Royal Astronomical Society*, 509(2):1521–1535, 2022. doi: 10.1093/mnras/stab2924.